

Challenge 8: Ryser's Hypergraph Conjecture

Definition 1 (r -partite). Given $r \in \mathbb{N}$, an r -uniform hypergraph $\mathcal{H}$ is r -partite if there is a partition

$$V(\mathcal{H}) = \bigcup_{i=1}^r P_i$$

of the vertex set of $\mathcal{H}$ such that $|P_i \cap e| = 1$, for each $e \in E(\mathcal{H})$ and $i \in [r]$.

Definition 2 (Cover number). Given a hypergraph $\mathcal{H}$, a set C of vertices of $\mathcal{H}$ is a *vertex cover* if it contains a vertex of every edge. The smallest cardinality of a vertex cover of $\mathcal{H}$ is called the *cover number* of $\mathcal{H}$. It is denoted by $\tau(\mathcal{H})$.

Definition 3 (Matching number). The *matching number* of a hypergraph $\mathcal{H}$ is the largest cardinality of a set of pairwise disjoint edges of $\mathcal{H}$. It is denoted by $\nu(\mathcal{H})$.

Conjecture 1.1 (Ryser's Conjecture). Let $r \in \mathbb{N}$. Every r -partite r -uniform hypergraph $\mathcal{H}$ satisfies:

$$\tau(\mathcal{H}) \leq (r - 1) \cdot \nu(\mathcal{H})$$

For $r = 2$, it is a classical theorem of König, and for $r = 3$ it was proved by Aharoni using the topological method.